CO1-AT1

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3. Impact of Frequency Reuse on cellular Network Performance.

FREQUENCY REUSE

1. Frequency planning
- Assigns frequencies to cells.
- Proper reuse distance
- Reduces interference

Leads to -> better signal quality

2. Co-channel Interface.
- same frequency in nearby cells.
- caused signal disturbance.
- Reduced SINR and call quality

Reduced by -> reuse distance

Reusing the same frequencies in different cells.

3. Adjacent channel Interface.
- Nearby frequency channels overlap.
- Causes unwanted signals.
- Degrades reception.

Reduced by -> guard bands and filtering.

4. Network capacity
- Frequency reuse increases capacity
- Supports more users.
- Excess interface reduces throughput

Improved by -> interface control.

INTERFERENCE REDUCTION TECHNIQUES

Power control . • sectoring • Directional antennas

• channel assignment • Filtering.

Conclusion :

• Frequency reuse increased network capacity by allowing the same spectrum to serve multiple cells.

• co-channel interference increases when cells using the same frequencies are too close.

• Adjacent channel interface is reduced through proper channel spacing, guard bands and filters.

• Interference reduction techniques improve signal quality and throughput, although sectoring and larger reuse distances can involve trade-offs in cost or capacity.